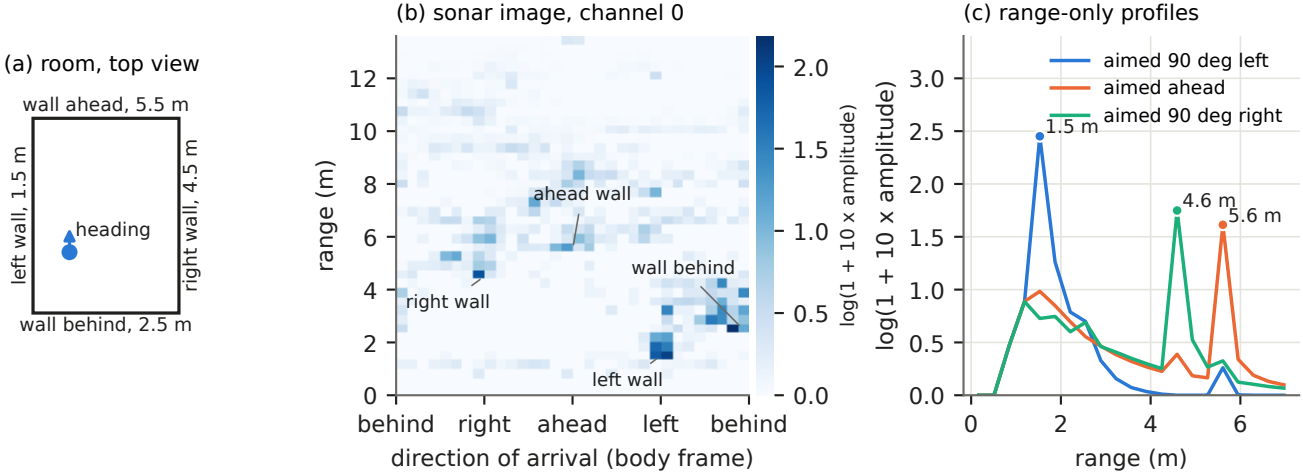
(a) room, top view
wall ahead, 5.5 m
left wall, 1.5 m
right wall, 4.5 m
heading
wall behind, 2.5 m
(b) sonar image, channel 0
range (m)
ahead wall
right wall
wall behind
left wall
behind
right
ahead
left
behind
direction of arrival (body frame)
log(1 + 10 x amplitude)
(c) range-only profiles
aimed 90 deg left
aimed ahead
aimed 90 deg right
1.5 m
4.6 m
5.6 m
log(1 + 10 x amplitude)
range (m)